KILLING A MAP FROM A PSEUDO-COHERENT COMPLEX BY A PERFECT TWIST THAT IS TRIVIAL OFF THE SUPPORT

THE STACKS PROJECT AUTHORS

Editorial note. Inputs lemma-quasi-coherence-internal-hom and lemma-generator-with-support are same-chapter NOT carded (the latter carries a reference block citing Rouquier-dimensions) and are declared external.

Allowed standard prerequisites (editorial).

- Quasi-coherence of the internal Hom. lemma-quasi-coherence-internal-hom (same chapter, NOT carded, declared external). Places K = R SheafHom(P, E) in D_QCoh and computes H^0(K) = Hom(O_X[0], K).
- Perfect generator with support. lemma-generator-with-support (same chapter, NOT carded, declared external). Provides a perfect generator E of the subcategory supported on T; its statement carries a reference block citing Rouquier-dimensions, hence not carded.
- Writing objects as homotopy colimits over a generator. Derived Categories, lemma write-as-colimit (cross-chapter, named). K = hocolim K_n with K_n built from the generator.
- Compactness of O_X and factorization. Derived Categories, lemmas commutes-with-countable-sums and factor-through (cross-chapter, named). The class of alpha lands in some K_n and factors through an object of the thick closure of the generator.

Source reference. Original reference 094A: focused-seeds/geometry/S9-perf2-map-from-pseudo-coherent-to-complex-with-support/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 09SP: focused-seeds/geometry/S9-perf2-map-from-pseudo-coherent-to-complex-with-support/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 09SN: focused-seeds/geometry/S9-perf2-map-from-pseudo-coherent-to-complex-with-support/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0A9A: sources/stacks-perfect/pinned-original-source.tex, source lines 4185–4185. Navigation pointer only: the counted proof

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cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0A6H: sources/stacks-perfect/pinned-original-source.tex, source lines 2657–2657. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let X be a quasi-compact and quasi-separated scheme. Let $T \subset X$ be a closed subset such that $U = X \setminus T$ is quasi-compact. Let $\alpha : P \to E$ be a morphism of $D_{QCoh}(\mathcal{O}_X)$ with either*

(1) *P is perfect and E supported on T, or*
(2) *P pseudo-coherent, E supported on T, and E bounded below.*

Then there exists a perfect complex of $\mathcal{O}_X$-modules I and a map $I \to \mathcal{O}_X[0]$ such that $I \otimes^{\mathbf{L}} P \to E$ is zero and such that $I|_U \to \mathcal{O}_U[0]$ is an isomorphism.

Proof. Set $\mathcal{D} = D_{QCoh,T}(\mathcal{O}_X)$. In both cases the complex $K = R\mathcal{H}om(P, E)$ is an object of $\mathcal{D}$. See Lemma 0A6H for quasi-coherence. It is clear that K is supported on T as formation of $R\mathcal{H}om$ commutes with restriction to opens. The map α defines an element of $H^0(K) = \mathrm{Hom}_{D(\mathcal{O}_X)}(\mathcal{O}_X[0], K)$. Then it suffices to prove the result for the map $\alpha : \mathcal{O}_X[0] \to K$.

Let $E \in \mathcal{D}$ be a perfect generator, see Lemma 0A9A. Write

$$K = \mathrm{hocolim} K_n$$

as in Derived Categories, Lemma 09SN using the generator E. Since the functor $\mathcal{D} \to D(\mathcal{O}_X)$ commutes with direct sums, we see that $K = \mathrm{hocolim} K_n$ holds in $D(\mathcal{O}_X)$. Since $\mathcal{O}_X$ is a compact object of $D(\mathcal{O}_X)$ we find an n and a morphism $\alpha_n : \mathcal{O}_X \to K_n$ which gives rise to α, see Derived Categories, Lemma 094A. By Derived Categories, Lemma 09SP applied to the morphism $\mathcal{O}_X[0] \to K_n$ in the ambient category $D(\mathcal{O}_X)$ we see that α_n factors as $\mathcal{O}_X[0] \to Q \to K_n$ where Q is an object of $\langle E \rangle$. We conclude that Q is a perfect complex supported on T.

Choose a distinguished triangle

$$I \to \mathcal{O}_X[0] \to Q \to I[1]$$

By construction I is perfect, the map $I \to \mathcal{O}_X[0]$ restricts to an isomorphism over U, and the composition $I \to K$ is zero as α factors through Q. This proves the lemma. □